MAHATMA GANDHI UNIVERSITY, KOTTAYAM
MGU-UGP (HONOURS)
FIFTH SEMESTER EXAMINATION
MODEL QUESTION PAPER
COURSE NAME: SPORTS NUTRITION ESSENTIALS
Course Code: MG5DSEPES305

Duration: 2 Hours **Maximum Marks: 70**

Part A

Multiple Choice Questions

Answer All Questions

Each question carries 1 mark (10 × 1 = 10)

1. Which of the following best describes the primary purpose of sports nutrition? [U, CO 1]
   a) To reduce the training frequency of athletes.
   b) To support health, performance and recovery through appropriate nutrition.
   c) To replace physical training with dietary strategies.
   d) To eliminate the need for hydration.
2. Which nutrient is the body's major source of energy during high-intensity exercise? [K, CO 2]
   a) Carbohydrate
   b) Protein
   c) Vitamins
   d) Minerals
3. Which of the following is an important principle of hydration before exercise? [U, CO 4]
   a) Begin exercise dehydrated.
   b) Avoid all fluids before exercise.
   c) Start exercise adequately hydrated.
   d) Drink only after exercise.
4. The main nutritional objective of a post-exercise meal is to promote: [U, CO 4,5]
   a) Dehydration
   b) Glycogen replenishment and muscle repair.
   c) Appetite suppression
   d) Reduction of recovery time by avoiding food.
5. Which nutrient has a central role in fuelling endurance activities? [K, CO 3]
   a) Carbohydrate
   b) Vitamin C
   c) Calcium
   d) Water

6. Which nutritional strategy is commonly associated with preparation for long-distance endurance events? [An, CO 3]

a) Carbohydrate loading

b) Complete fasting

c) Avoiding fluids

d) Eliminating carbohydrates

7. Which approach is most appropriate for healthy weight loss in an athlete? [E, CO 3,4]

a) Severe calorie restriction

b) A moderate and sustainable energy deficit.

c) Skipping meals regularly

d) Eliminating all dietary fat

8. Which practice is most appropriate for healthy weight gain while maintaining performance? [A, CO 3]

a) A moderate, controlled energy surplus

b) Prolonged fasting

c) Rapid excessive overeating

d) Avoiding protein

9. Which of the following is an unhealthy weight-control practice that athletes should avoid? [An, CO 3,4]

a) Monitoring body composition appropriately

b) Gradual dietary adjustment

c) Severe restriction and rapid weight loss

d) Seeking professional guidance

10. Which psychological factor can influence athletic weight management? [U, CO 3,4]

a) Body image pressure

b) Muscle glycogen only

c) Fluid temperature only

d) Meal colour

Part B

Short Answer Type Questions

Answer any 5 Questions

Each question carries 2 marks (5 × 2 = 10)

11. Define sports nutrition and state its importance in athletic performance. [U, CO 1]

12. List any four essential nutrients and mention one role of each. [K, CO 2]

13. What are the general principles of hydration before and during exercise? [U, CO 1,4]

14. What is pre-exercise nutrition? Mention any two factors affecting meal timing. [U, CO 4,5]

15. What is glycogen replenishment and why is it important after exercise? [K, CO 2,5]

16. State any four nutritional requirements of endurance athletes. [K, CO 3]

17. What are the psychological factors that influence weight management in athletes? [An, CO 3,4]

Part C

Short Essay Type Questions

Answer any 4 Questions

Each question carries 5 marks (4 × 5 = 20)

18. Explain the principles of sports nutrition and discuss their significance in athletic performance. [U, CO 1,3]
19. Explain the timing and composition of meals before training and competition. [U, CO 4,5]
20. Discuss nutritional considerations for optimal recovery, including glycogen replenishment, muscle repair and rehydration. [An, CO 2,4,5]
21. Explain the specialized nutritional needs of endurance athletes and the major fuelling strategies used during long-distance events. [An, CO 3]
22. Describe the nutritional requirements and dietary strategies for strength and power activities. [U, CO 2,3]
23. Explain healthy approaches to weight loss and weight gain while maintaining athletic performance. [E, CO 3,4]

Part D

Essay Type Questions

Answer any 2 Questions

Each question carries 15 marks (2 × 15 = 30)

24. Explain in detail the principles and significance of sports nutrition. Discuss the classification, roles and sources of nutrients and explain hydration strategies before, during and after exercise. [K, U, CO 1,2,4]
25. Discuss pre- and post-exercise nutrition in detail. Explain meal timing and composition, balancing energy needs with digestive comfort, glycogen replenishment, muscle repair, rehydration and practical recovery strategies. [A, An, CO 4,5]
26. Explain the nutritional requirements of endurance, strength and power athletes. Discuss carbohydrate fuelling, protein needs, hydration, competition-day nutrition, muscle development and recovery. [U, An, CO 2,3]
27. Explain healthy weight management strategies for athletes. Discuss weight loss, weight gain, evaluation of body composition, unhealthy weight-control practices, low energy availability, psychological factors, and the role of the coach/trainer in supporting a healthy mindset. [E, An, CO 3,4]